

COMP64301 Coursework

Yiming Fu

MSc Artificial Intelligence

The University of Manchester

Manchester, United Kingdom

yiming-fu-2@postgrad.manchester.ac.uk

Abstract—This work takes image classification as a representative task to systematically examine the evolution and emerging trends of computer vision techniques in robotics. To better understand the differences and commonalities between traditional vision pipelines and CNN-based models, we design a lightweight 4-layer CNN and compare it with the ResNet architecture, further analyzing the advantages of transfer learning in small-sample settings. For the traditional methods, we conduct ablation studies across local descriptors (SIFT, ORB), the global descriptor HOG, and their combined configurations to evaluate trade-offs in accuracy and computational complexity. Using the iCub and STL-10 datasets, we assess model generalization under background changes and investigate performance in challenging real-world scenes. Confusion matrices and feature visualizations provide deeper insights into the behavior of traditional and CNN-based approaches. Finally, we discuss the limitations of our experimental design in relation to current state-of-the-art developments in visual perception for robotics.

Index Terms—Image Classification, Robotics.

I. INTRODUCTION

Computer vision plays a central role in robotic perception, enabling robots to interpret the physical world and make autonomous decisions based on visual information. Foundational CV tasks such as image classification, generic object detection, and semantic segmentation provide essential capabilities for object recognition, semantic scene understanding, obstacle localization, and graspable-region analysis within robotic environments [1], [2]. When integrated with Visual SLAM, depth estimation, 3D pose estimation, and visual servoing, these perception outputs establish a structured understanding of the environment, serving as critical inputs for downstream modules including task planning, grasp strategy design, and interactive decision-making [3], [4]. Moreover, the fusion of visual models with multimodal sensors has become increasingly important [5], [6]. Which can enhance robustness and enable robots to operate effectively in complex, dynamic, and unstructured environments.

A. Trends for Computer Vision in Robotics

Driven by rapid advancements in deep learning, the role of computer vision in robotics is shifting from modular perception to end-to-end embodied intelligence. Traditional robotic architectures typically adopt a layered pipeline in which vision operates as an independent front-end module, relying on localization algorithms such as the ORB-SLAM family [3] and detectors including YOLO [7] and Mask R-CNN [2] to extract state information before passing it to downstream planning

and control. However, recent progress in 2023–2025 positions vision as the core enabler of physical-world interaction. In environmental representation, 3D Gaussian Splatting (3DGS) has emerged as a powerful explicit radiance-field technique, gradually replacing conventional point-cloud and mesh-based pipelines by offering high-fidelity, real-time, and semantically enriched scene reconstruction. In decision-making and control, Transformer-based Vision–Language–Action (VLA) foundation models, such as RT-2 [8] and Octo [9], together with Diffusion Policy approaches have begun to dissolve the traditional boundaries between perception and control. By leveraging large-scale multimodal training, these methods enable direct pixel-to-action mappings, empowering robots to operate through unified, end-to-end visuomotor intelligence with enhanced adaptability and generalization.

B. Image Classification in Robotics

Image classification is one of the crucial parts of vision perception in robotics. Its applications span from operational domain awareness (distinguishing between indoor, outdoor, and industrial workshop environments) to fine-grained manipulation tasks such as identifying graspable objects, and safety-critical scenarios in chemical industries requiring the precise recognition of hazardous materials or specialized warning insignias [10].

In early computer vision methods, feature extraction relied predominantly on hand-crafted descriptors. Techniques such as Scale-Invariant Feature Transform (SIFT) [11] and Histogram of Oriented Gradients (HOG) [12], combined with Support Vector Machines (SVMs) or sliding-window detectors, demonstrated stability in controlled, static settings. However, these data-independent methods lack the capacity to model complex semantic representations, resulting in poor generalization when faced with real-world challenges such as drastic illumination changes, object occlusions, and viewpoint deviations.

In recent years, the introduction of Convolutional Neural Networks (CNNs) marked a paradigm shift, significantly enhancing the semantic expressiveness of visual features through hierarchical representation learning. Simonyan et al. proposed VGGNet [13], which demonstrated that increasing network depth using small convolutional filters significantly enhances the representation capability for complex object categories. Subsequently, He et al. introduced ResNet [14], utilizing residual skip connections to solve the vanishing gradient problem in deep networks, allowing robots to leverage ultra-

deep architectures for high-precision recognition. To address the stringent Size, Weight, and Power (SWaP) constraints in mobile robotics, Tan et al. developed EfficientNet [15], which introduced a compound scaling method to optimize accuracy under limited computational budgets, becoming a standard backbone for edge-deployed perception systems.

Despite their success, CNN-based methods inherently suffer from limited local receptive fields, which hinders the effective modeling of global semantic relationships essential for complex scene understanding. To address this limitation, Dosovitskiy et al. proposed the Vision Transformer (ViT) [16], introducing a global self-attention mechanism. This architecture effectively captures long-range dependencies within images, thereby deepening the understanding of complex scene structures. Subsequently, Radford et al. introduced CLIP [17], which aligned visual and linguistic latent spaces through large-scale contrastive learning. This breakthrough empowered robots with open-vocabulary perception, enabling agents to recognize novel objects and interpret complex semantic instructions beyond predefined training categories.

In the 2023–2025 landscape, visual perception is converging with Large Language Models (LLMs) to form Vision-Language-Action (VLA) architectures (e.g., RT-2 [8]), enabling robots to perform continuous learning from unlabeled data. To address the computational bottlenecks of Transformers, Gu et al. introduced Mamba [18], a hardware-aware Selective State Space Model (SSM) that achieves linear inference efficiency while retaining global context modeling. Building on this, Zhu et al. proposed Vision Mamba (Vim) [19], which adapts bidirectional SSMs for vision tasks. Vim demonstrates superior efficiency, achieving $2.8\times$ faster inference than ViT-DeiT on edge devices, significantly reducing latency for high-speed robotic maneuvers. Furthermore, focusing on the challenge of Test-Time Adaptation (TTA) in dynamic environments, Wang et al. proposed CoTTA [20], a continual test-time adaptation method that allows classification models to adjust to changing weather or lighting conditions online without catastrophic forgetting, ensuring robust deployment in the wild.

II. METHODS

A. Dataset

To allow a fair comparison of different image classification approaches under varying background conditions, changes in the object holder, and real-world image degradations, we employ two complementary datasets: iCubWorld 1.0 and STL-10.

The iCubWorld 1.0 dataset is collected using the visual sensors of the iCub humanoid robot, where a human demonstrator presents objects in an indoor laboratory environment. It contains ten object categories, with RGB images of size 160×160 . The test set includes several subcategories such as *demonstrator* and *robot* views, which are designed to evaluate model generalization under category-level variations (e.g., different instances within the same class) and to assess

the impact of changing the object holder and background when shifting from a human to a robot perspective.

The STL-10 dataset, derived from CIFAR-10, comprises ten outdoor object categories with RGB images of size 96×96 , and additionally provides a large set of unlabeled images for unsupervised learning. Its real-world characteristics such as multiple objects per scene, object stacking, motion blur, and complex background clutter, introduce significant challenges for image classification models.

B. Data Preprocessing

To ensure a controlled comparison across datasets, we align the data scale by selecting 10 categories with 400 training images, 100 validation images, and 200 test images per category for both datasets. Because image resolution differences can substantially influence classification performance, particularly for traditional feature extraction methods. All images are resized to a unified and commonly adopted resolution of 128×128 . This resolution strikes a balance by preserving essential structural information from both datasets while avoiding artifacts introduced by excessive upscaling or downscaling.

To introduce controlled appearance variation while preserving the datasets' inherent characteristics, we apply a consistent augmentation pipeline composed of a random resized crop (scale range 0.8–1.0), random horizontal flipping, random rotation of 15° , and color jittering in brightness, contrast, saturation, and hue. The processed images are then converted to tensors and normalized using the specified mean and standard deviation values.

C. Algorithm

1) *Traditional Algorithms*: For traditional, non-neural network-based computer vision methods, Canny edge detection [22] and Harris corner detectors [23] are among the most widely used techniques for extracting low-level geometric features. Although they effectively capture basic structural information, these methods are limited by semantic ambiguity and high sensitivity to environmental noise.

To more effectively address the illumination and background variations commonly encountered in robotic image classification tasks, we conducted a comparative evaluation of three representative handcrafted descriptors. The Scale-Invariant Feature Transform (SIFT) [24] was included due to its design emphasis on robustness to scale and rotation changes, which be considered particularly beneficial for handling variable viewpoints in robotic perception. The Histogram of Oriented Gradients (HOG) [25] was selected as a global gradient-based descriptor capable of capturing coarse object structure under moderate appearance variations. ORB [26], in contrast, was chosen for its computational efficiency and binary representation, making it well suited for real-time robotic applications with limited onboard processing resources.

In this study, we compared these three feature extraction techniques and additionally explored hybrid strategies that combine local descriptors (SIFT/ORB) with global features

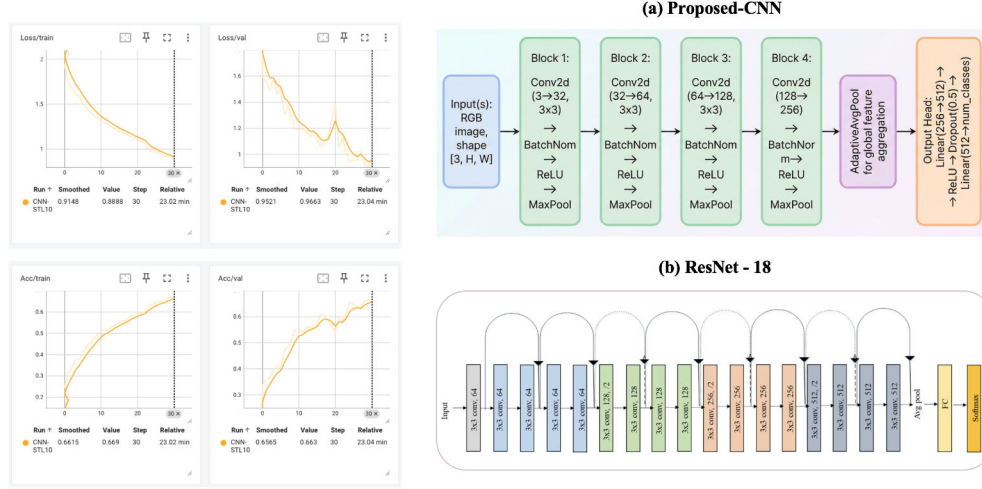


Fig. 1: Training performance and model architectures. Left: Training and validation curves for Loss and Accuracy on the STL-10 dataset over 30 epochs. Right: (a) The detailed architecture of the Proposed-CNN, consisting of four sequential convolutional blocks and a classification head. (b) The schematic diagram of the ResNet-18 structure (adapted from [21]).

TABLE I: Performance Comparison of Handcrafted Feature Extraction Methods in Icub Dataset

Exp. No.	Method	Accuracy (%)	Time (ms)
1	SIFT	20.44	32.99
2	HOG	33.98	64.30
3	ORB	17.74	17.92
4	SIFT + HOG	32.98	126.43
5	ORB + HOG	35.23	51.24

(HOG) to examine their trade-offs in accuracy and computational cost under different environmental conditions, as summarized in Section III-A. To address the varying lengths of local feature descriptors, the Bag of Words (BoW) framework [27] was employed to convert keypoint-based features into fixed-size visual vocabularies. Furthermore, Support Vector Machines (SVM) [28] were selected as the primary classifier due to their strong performance in high-dimensional feature spaces and well-established effectiveness in traditional vision pipelines.

2) *CNN-based Algorithms*: Considering that the original training dataset is relatively small (4,000 samples across 10 classes), and that the images have low resolution and contain only a single object mostly, the model design in this study prioritizes a balance between parameter efficiency and feature extraction capability, while avoiding overly deep or structurally complex networks. Following this principle, we construct a lightweight four-layer convolutional neural network. The network consists of four convolutional blocks, each comprising a 3×3 convolution, batch normalization (BN), and a ReLU activation, followed by max pooling to progressively reduce the spatial resolution and enlarge the receptive field. After the four convolutional blocks, an adaptive global average

pooling layer is applied to compress the feature maps into channel-level representations, which are then passed to a two-layer fully connected classifier with dropout regularization. This structure maintains sufficient representational capacity while effectively mitigating overfitting, making it well suited for low-resolution, small-sample image classification tasks. The detailed network architecture is illustrated in Fig. 1.

To comprehensively evaluate the effectiveness of the proposed lightweight CNN and examine how increased network depth influences performance under low-resolution and limited-sample conditions, we introduced ResNet18 as a comparative baseline, which employs residual connections to stabilize the training of deeper models and is widely used as a generic feature extractor. In our experiments, we included both a version trained from scratch, which tests the ability of a deep network to learn directly from the target data without prior knowledge, and a transfer learning variant initialized with ImageNet pretrained weights.

D. Hyperparameter Exploration Strategy

1) *Traditional Method*: For the classical, handcrafted feature-based approach, the hyperparameter exploration focuses on a limited set of key parameters that directly influence feature representation and classifier performance. Specifically, for ORB-based local descriptors, the number of keypoints per image is explored across $\{300, 500, 700\}$, and the size of the visual vocabulary in the Bag-of-Words (BoW) model is varied in $\{500, 1000, 1500\}$, chosen to balance descriptive power with robustness given the limited dataset size and the relatively small number of training samples per class. For HOG descriptors, the pixel-per-cell parameter is tested at multiple scales ($8 \times 8, 16 \times 16, 32 \times 32$) to capture both fine-grained and coarse structural patterns relevant for object

TABLE II: Results of Hyperparameter Tuning for Traditional Methods: Accuracy and Computational Time.

BOW	ORB	HOG	Performance	
			Accuracy	Time(s)
	300	8	0.3073	322.62
		16	0.3538	243.72
		32	0.3503	101.30
	500	8	0.3073	293.51
		16	0.3548	42.15
		32	0.3508	18.19
	700	8	0.3073	269.27
		16	0.3548	39.57
		32	0.3508	17.68
	300	8	0.3078	1707.94
		16	0.3528	340.65
		32	0.3493	140.48
	500	8	0.3078	278.52
		16	0.3523	51.25
		32	0.3518	22.41
	700	8	0.3078	259.71
		16	0.3523	51.31
		32	0.3518	22.63
	300	8	0.3083	853.82
		16	0.3523	63.45
		32	0.3468	28.97
	500	8	0.3088	295.28
		16	0.3513	63.13
		32	0.3478	26.45
	700	8	0.3088	288.56
		16	0.3513	58.02
		32	0.3478	27.62

retrieval in robotics. These feature parameters are combined in a grid-like fashion, generating a manageable number of configurations for evaluation.

2) *CNN-based Method*: To avoid the opacity and excessive computational cost associated with AutoML-based hyperparameter tuning, we adopt a constrained and interpretable exploration strategy grounded in the scale of the available data, the chosen model capacity, and the characteristics of the robotic perception task. Accordingly, the model is trained using the Adam optimizer [29] with an initial learning rate of 1×10^{-3} , a batch size of 16 to preserve sufficient gradient diversity under limited samples, and a weight decay of 1×10^{-4} combined with a dropout rate of 0.5 [30] to mitigate overfitting. To ensure robust convergence, a dynamic learning-rate scheduler [31] is employed, reducing the learning rate by a factor of 0.5 when validation accuracy stagnates for five consecutive epochs. Furthermore, to prevent over-training and conserve computational resources, the training process incorporates an early stopping mechanism [32]. The maximum epoch budget is set to 50, but training terminates automatically if no improvement in validation accuracy is observed for ten consecutive epochs. This approach ensures the retention of the optimal model parameters, effectively balancing bias and variance under small-sample conditions.

III. EXPERIMENT RESULTS

All experiments were conducted on a workstation running Ubuntu 20.04.6 with an Intel Core i9 CPU and an NVIDIA

RTX 4060 GPU with 8 GB of memory. CUDA 12.2 was used for GPU acceleration.

A. Training and Results

1) *Traditional Method*: As shown in Table I, we compared local feature extraction methods (SIFT and ORB), the global feature HOG, and their combinations. In both the BoW and SVM stages, the same hyperparameter configurations were used. The results indicate that ORB achieves superior feature extraction speed (17.92s vs. 32.99s for SIFT), outperforming other feature extraction methods. Furthermore, combining ORB with the global HOG feature significantly improves classification accuracy, yielding a 17.49% increase compared to using ORB alone. This improvement can be attributed to the complementary nature of the features, where ORB captures local keypoint details and HOG encodes global shape and structure. Since the original dataset was captured in a controlled laboratory environment with limited rotation and viewpoint variation, SIFT did not demonstrate a noticeable advantage during feature extraction process.

Table II summarizes the results of hyperparameter tuning for the traditional feature-based methods. Accuracy remains relatively stable across different ORB keypoint numbers (300, 500, 700), indicating that increasing the number of local keypoints yields limited improvement given the available training samples. The HOG pixel-per-cell parameter has a more pronounced effect. A medium scale of 16×16 achieves the highest accuracy (0.3548), balancing fine-grained detail capture with robustness. Larger BoW vocabulary sizes (1000–1500) do not improve accuracy compared to BoW=500, while slightly increasing computational cost for the optimal HOG settings. Consequently, the configuration with BoW=500, ORB=500, and HOG= 16×16 yields the best classification performance. Alternatively, using HOG= 32×32 offers a viable trade-off, reducing computational time by over 50% (from 42.15s to 18.19s) with only a marginal drop in accuracy.

2) *CNN-based Method*: Following the hyperparameter exploration strategy outlined in Section II-C, we evaluated the proposed CNN method alongside the ResNet-18 framework. We investigated two distinct initialization settings for ResNet: transfer learning with pre-trained weights and training from scratch. To ensure a fair comparison, we identified the optimal hyperparameter combinations for each method on both the iCub and STL-10 datasets. The comparative results are presented in Fig. 1. Taking the training performance of the CNN model on the STL-10 dataset as a representative case, both the training and validation losses demonstrate a steady and consistent downward trend. This convergence indicates that the model has effectively learned features with good generalization capabilities.

As shown in Table III, for the small-scale iCub and STL-10 dataset (10 classes with only 40 training images per class), deeper architectures such as ResNet-18 do not exhibit a clear performance advantage over the proposed lightweight four-layer CNN. On the single-scene classification task, the CNN achieves an accuracy of 55.77%, outperforming ResNet-18

TABLE III: Comparison of Model Complexity, Efficiency, and Accuracy Across Datasets

Model	Parameters (M)	FLOPs (G)	Average Inference (ms)	Accuracy (%)			
				iCub Categorization	iCub Demonstrator	iCub Robot	STL-10
ORB + HOG	—	—	—	35.33	23.74	9.85	59.00
Proposed-CNN	8.78	0.253	2.28	55.77	51.82	18.83	30.25
ResNet18 (Transfer Learning)	11.18	1.824	15.52	78.21	63.62	50.12	90.65
ResNet18 (Scratch)	11.18	1.824	13.48	43.38	37.98	11.72	44.40

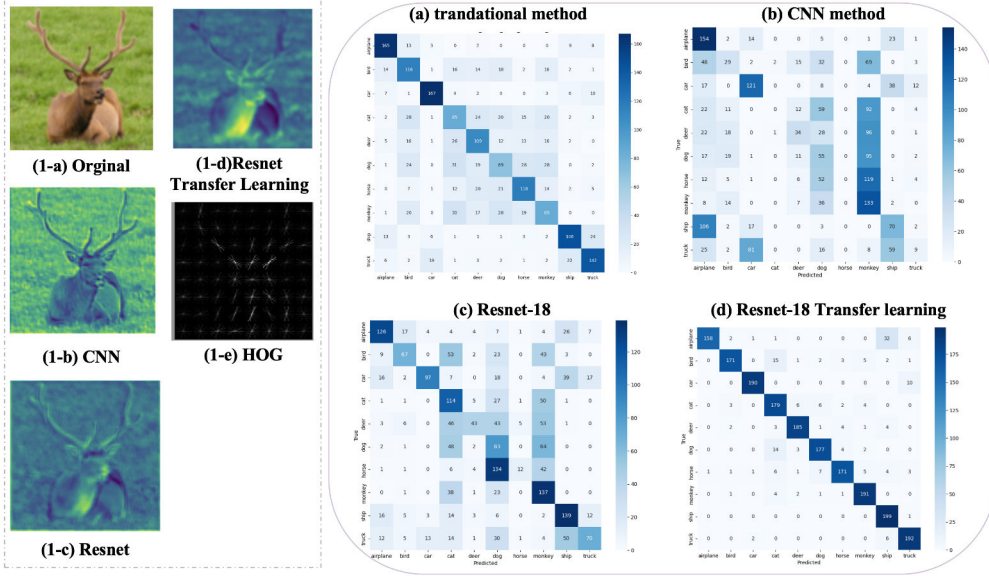


Fig. 2: Feature and performance visualization on the STL-10 dataset. Left panel: Comparison of feature extraction capabilities showing (1-a) the original input, (1-b, 1-c) intermediate feature maps of CNN and ResNet, (1-d) the attention heatmap of ResNet with Transfer Learning, and (1-e) HOG features. Right panel: Confusion matrices comparing classification accuracy among (a) Traditional method, (b) Custom CNN method, (c) Standard ResNet-18, and (d) ResNet-18 with Transfer Learning.

trained from scratch (43.38%). This outcome is expected, as the limited amount of training data is insufficient to fully optimize the substantially larger parameter space of ResNet-18. Transfer learning, however, provides substantial benefits in small-sample cases. On the iCub dataset, introducing pre-trained weights increases accuracy by nearly 35%. The robot-view evaluation set in iCub which changes the viewpoint from human to robot and simultaneously alters the background, which serves as a useful tool for assessing whether the model relies on background cues. Under this more challenging test condition, the ResNet-18 with transfer learning model achieves 50.12% accuracy, significantly outperforming the CNN and the scratch-trained ResNet-18 models (18.83% and 11.72%, respectively). For the STL-10 dataset, the transfer learning configuration again leads by a large margin, achieving 90.65% accuracy. This strong result is consistent with the fact that STL-10 is derived from CIFAR-10 and may share distributional similarities with the images used in the ResNet-18

pretraining corpus. In contrast, the CNN and the scratch-trained ResNet-18 perform slightly worse on STL-10 than on the iCub dataset, likely due to STL-10’s more complex outdoor scenes, richer backgrounds, and occasional motion blur.

B. Results Comparison

As shown in the STL-10 confusion matrix in Fig. 2, the ORB+HOG feature extraction pipeline achieves a clearly superior classification accuracy (59%) compared with the proposed CNN model (30.25%) and the ResNet-18 trained from scratch (44.40%). This accuracy gap can be explained by the inherent difficulty of the STL-10 dataset, which contains complex outdoor scenes with object occlusion, motion blur, and cluttered backgrounds. As illustrated in the confusion matrix, both the CNN and the non-pretrained ResNet-18 frequently confuse animal classes, particularly *deer* and *horse*. In contrast, the traditional ORB+HOG approach achieves significantly better results on these categories. Feature visualizations further support this observation. HOG descriptors capture strong edge and

shape cues for objects such as *deer*, providing discriminative structure that facilitates downstream classification. Meanwhile, the CNN-based models, trained with relatively limited samples under challenging visual conditions, shows weak or diffuse activation patterns, indicating that they struggle to learn stable, object-centric features from the available data.

This finding suggests that even in an era dominated by Transformers and vision-language models (VLMs), traditional computer vision techniques still hold an irreplaceable position. Owing to their higher interpretability and lower computational complexity, these methods continue to play an important role in image classification tasks within specialized scenarios such as small or zero-shot industrial inspection [33] and computation-constrained micro-robotic systems.

LIMITATIONS AND FUTURE WORK

This study has several limitations that point to promising directions for future research. First, due to constraints in time and computational resources, cross-validation was not employed during model training. Cross-validation is particularly important in small-sample settings, as it provides more reliable estimates of generalization performance and reduces sensitivity to dataset partitioning. Although the training and validation curves indicate stable learning without signs of overfitting, further hyperparameter refinement and extended training epochs may improve performance, especially for the CNN-based models.

For the traditional feature-based pipeline, the comparison was limited to three representative configurations involving SIFT, ORB, and HOG. Given the relatively low input resolution (128×128), local descriptors may suffer from feature overlap and repeated keypoint responses, potentially limiting discriminative power. Future work may explore multi-scale keypoint selection, enhanced Bag-of-Words variants or the application of sparse coding and dictionary learning techniques.

Regarding model architecture design, we did not perform fine-grained exploration of network depth, width, or residual pathways. Incorporating lightweight residual blocks or bottleneck structures may offer a better trade-off between model capacity and parameter efficiency. Furthermore, analysis of the iCub “robot-view” subset reveals that the current models struggle to disentangle background variations from object-centric cues, suggesting limited robustness to context shifts.

Future extensions may consider Vision Transformers (ViTs), which exhibit stronger global receptive fields and improved robustness to background bias. Beyond this, post-2023 advances such as CLIP-style vision-language pretraining and VLM decoders provide promising avenues for addressing fine-grained classification and open-vocabulary recognition in realistic, dynamic environments. Such models may ultimately bridge the gap between low-level perception and semantic-level reasoning required for robotic autonomy.

IV. CONCLUSION

In this work, we use image classification as a point to examine the development and application of computer vision tech-

niques in the robotics domain. By comparing classical computer vision methods with CNN-based approaches, we find that CNNs generally demonstrate superior performance, with transfer learning proving particularly beneficial for small-sample training scenarios. Nonetheless, traditional methods maintain their relevance in certain tasks due to their interpretable feature extraction schemes, lower computational requirements, and, in some cases, competitive accuracy. This makes them suitable for downstream applications with strict real-time or low-resource constraints, such as industrial robotics. Furthermore, by analyzing training parameter outcomes, confusion matrices, and feature visualizations, we summarize the limitations of the present study and outline directions for future research.

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